

Orbital Rotation Gradient and Hessian for a General JK-only Functional

Setup

Fixed natural occupations n_p , $D_{pq} = n_p \delta_{pq}$. A JK-only two-electron energy functional with a general symmetric occupation-coupling function $f(n_a, n_b) = f(n_b, n_a)$, $f(n, n) = n$, replacing the two-particle density matrix elements as

$$\Gamma_{pqpq} = \frac{1}{2} n_p n_q \quad (\text{Hartree/direct}), \quad \Gamma_{pqqp} = -\frac{1}{2} f(n_p, n_q) \quad (\text{exchange}), \quad (1)$$

so that

$$E_2 = \frac{1}{2} \sum_{pq} n_p n_q \langle pq|pq \rangle - \frac{1}{2} \sum_{pq} f(n_p, n_q) \langle pq|qp \rangle. \quad (2)$$

The Müller functional is the special case $f(n_a, n_b) = \sqrt{n_a n_b}$.

Important: this Γ does *not* satisfy the fermionic antisymmetry $\Gamma_{pqrs} = -\Gamma_{qprs}$ of a genuine two-particle density matrix (it is an N -representability-violating reconstruction, valid only in the idempotent/integer-occupation limit). The gradient and Hessian below are therefore derived directly from the explicit energy expression above (differentiating each integral leg under orbital rotation), *not* from the operator-commutator machinery used for the general Γ case, since that machinery relied on antisymmetry at two separate steps.

Gradient

Define

$$V_{qp} \equiv h_{qp} + \sum_t n_t \langle tq|tp \rangle. \quad (3)$$

$$\boxed{g_{pq} = (n_q - n_p) V_{qp} + \sum_t [f(n_p, n_t) - f(n_q, n_t)] \langle qt|tp \rangle} \quad (4)$$

Hessian

$$\begin{aligned} G_{pq,rs} = & (n_q - n_p) \left[-\delta_{qr} V_{sp} + \delta_{ps} V_{qr} + (n_s - n_r) \langle sq|rp \rangle \right] \\ & - \delta_{qr} \sum_t [f(n_p, n_t) - f(n_r, n_t)] \langle st|tp \rangle + \delta_{ps} \sum_t [f(n_s, n_t) - f(n_q, n_t)] \langle qt|tr \rangle \\ & + [f(n_p, n_s) - f(n_p, n_r) - f(n_q, n_s) + f(n_q, n_r)] \langle qs|rp \rangle \end{aligned} \quad (5)$$

Special case (Müller): set $f(n_a, n_b) = \sqrt{n_a n_b}$ everywhere above.

Generalization: independent Hartree coupling

The derivation only uses that the couplings are fixed, symmetric matrices attached to orbital positions, so the Hartree factor $n_p n_q$ may be replaced by an arbitrary symmetric H_{pq} (needed for the antisymmetric Müller variant, whose Hartree coupling is $g(n_p, n_q) = \frac{1}{2}[n_p n_q - \sqrt{n_p n_q}]$, equal to its exchange coupling), and $f(n_p, n_q)$ by a symmetric X_{pq} (index-dependent couplings such as BBC2 or the diagonal branch of MLSIC/GU need nothing else). With $E = \sum_a n_a h_{aa} + \frac{1}{2} \sum_{ab} H_{ab} \langle ab|ab \rangle - \frac{1}{2} \sum_{ab} X_{ab} \langle ab|ba \rangle$, the same leg-by-leg differentiation gives

$$g_{pq} = (n_q - n_p) h_{qp} + \sum_t \left\{ [H_{qt} - H_{pt}] \langle tq|tp \rangle + [X_{pt} - X_{qt}] \langle qt|tp \rangle \right\}, \quad (6)$$

$$\begin{aligned} G_{pq,rs} = & (n_q - n_p) [-\delta_{qr} h_{sp} + \delta_{ps} h_{qr}] - \delta_{qr} \sum_t [H_{qt} - H_{pt}] \langle ts|tp \rangle + \delta_{ps} \sum_t [H_{qt} - H_{pt}] \langle tq|tr \rangle \\ & - \delta_{qr} \sum_t [X_{pt} - X_{qt}] \langle st|tp \rangle + \delta_{ps} \sum_t [X_{pt} - X_{qt}] \langle qt|tr \rangle \\ & + [H_{qs} - H_{qr} - H_{ps} + H_{pr}] \langle sq|rp \rangle + [X_{ps} - X_{pr} - X_{qs} + X_{qr}] \langle qs|rp \rangle. \end{aligned} \quad (7)$$

For $H_{ab} = n_a n_b$ these are exactly the boxed gradient and the Hessian above (with V_{qp} expanded and $f \rightarrow X$), since $(n_q - n_p)(n_s - n_r) = H_{qs} - H_{qr} - H_{ps} + H_{pr}$.

Real-step Hessian and code conventions

$G_{pq,rs}$ is the “bare” coefficient, the derivative of g_{pq} along a rotation generated by κ_{rs} (a sequential rotation, which is what a finite difference of the gradient at a rotated point measures). The Hessian for the real orbital-rotation step $\kappa_{pq} = +t$, $\kappa_{qp} = -t$ is

$$\text{Hess}_{pq,rs} = \frac{\partial^2 E}{\partial \kappa_{pq} \partial \kappa_{rs}} = G_{pq,rs} - G_{pq,sr} - G_{qp,rs} + G_{qp,sr}, \quad (8)$$

as in `orbital_hessian.tex`. In the code (`Hessian_opt/JkOnlyFock.h`, `JkOnlyHessian.h`) the gradient is stored with the project convention $g_{pq}^{\text{code}} = -2 g_{pq}^*$ (only $p \geq q$; $\text{Re } g_{pq}^{\text{code}} = \partial E / \partial \kappa_{pq}$), and `jkOnlyHessianElement` returns $\text{Hess}_{pq,rs}$ (its imaginary part is round-off for physical input).

Numerical validation

Status. The gradient and Hessian above have been checked numerically against finite differences, for every JK-only functional implemented in the code (SD, Müller, BBC2, CA, CGA, ML, MLSIC, GU, POWER, and the antisymmetric Müller variant), for real orbitals (nonrelativistic) and complex spinors (X2C and four-component Dirac–Hartree–Fock). They were also re-derived independently by differentiating the rotated integral legs (as in the generalization above); this reproduces the boxed expressions term by term.

Test protocol. Orbitals are rotated by $e^{-\kappa}$ with $\kappa_{rs} = +t$, $\kappa_{sr} = -t$ (one-electron and two-electron integrals transformed accordingly), at *fixed* occupations, with central steps $t = \pm 10^{-4}$.

- *Gradient:* $\text{Re } g_{pq}^{\text{code}}$ against $[E(+t) - E(-t)]/2t$, with E the energy above at the rotated integrals; and elementwise against the gradient obtained from the generalized Fock matrix (an independent route).

- *Hessian*: $\text{Hess}_{pq,rs}$ against $[g_{pq}(+t) - g_{pq}(-t)]/2t$, where $g_{pq}(\pm t)$ is the (independently validated, Fock-matrix based) gradient evaluated at the rotated integrals.

Example 1: all functionals at generic fractional occupations. Water, geometry (Å) O (0, 0, 0), H (0, ± 0.757 , 0.586), STO-3G basis, 10 electrons. Occupations of the active orbitals (all spin-orbitals/spinors of the positive-energy space) were fixed to the generic values $n_i = 0.15 + 0.7 \{0.6180339887(i + 1)\}$ ($\{\cdot\}$ the fractional part), i.e. strictly inside (0, 1) and *not* the result of any optimization, so that no functional is tested in a trivial near-idempotent limit. The three orbital pairs with the largest gradient were used automatically; with the two orientations of (r, s) this gives 18 index quadruples (p, q, r, s) per functional, including ones sharing an index. Table 1 lists the largest deviations for the X2C spinors (for the other two cases see below). Here $\max |\Delta g|$ is the deviation of the gradient from the energy finite difference, $\max |\Delta \text{Hess}|$ that of the Hessian from the finite difference of the gradient (the Hessian elements have magnitudes up to ~ 1). For real

functional	$\max \Delta g $	$\max \Delta \text{Hess} $
SD (HF)	4.6×10^{-9}	5.7×10^{-9}
Müller	4.7×10^{-9}	6.2×10^{-9}
BBC2	5.0×10^{-9}	5.6×10^{-9}
CA	4.7×10^{-9}	5.6×10^{-9}
CGA	4.7×10^{-9}	5.8×10^{-9}
ML	4.6×10^{-9}	5.7×10^{-9}
MLSIC	4.6×10^{-9}	6.1×10^{-9}
GU	4.7×10^{-9}	1.2×10^{-8}
POWER ($\alpha = 0.6$)	4.7×10^{-9}	5.8×10^{-9}
antisym. Müller	1.1×10^{-8}	8.1×10^{-9}

Table 1: X2C spinors, water/STO-3G, generic fractional occupations, 18 quadruples per functional. Deviations of the analytic gradient (vs. energy finite difference) and Hessian (vs. finite difference of the gradient), consistent with the $O(t^2)$ truncation error of the central difference.

orbitals (nonrelativistic) the deviations are at most 8×10^{-9} (gradient) and 1.2×10^{-8} (Hessian), and for four-component spinors 9×10^{-9} and 1.9×10^{-8} . The gradient agrees elementwise with the Fock-matrix gradient to $\leq 8 \times 10^{-13}$, and the imaginary part of the Hessian is $\lesssim 10^{-15}$.

Example 2: Müller at its own optimized occupations. Same molecule and basis, $f = \sqrt{n_a n_b}$, occupations optimized at fixed orbitals (all occupations 1.0 except the frontier ones; the pair below has $n_p \approx 0.836$, $n_q \approx 0.094$, both genuinely fractional, with a nonzero gradient). For the pair $(p, q) = (8, 10)$ and $(r, s) = (9, 11)$ (0-based orbital indices):

	analytic	finite difference	diff
$\text{Hess}_{pq,rs}$, X2C	0.02017414077	0.02017414064	1.3×10^{-10}
$\text{Hess}_{pq,rs}$, 4-component	0.004022140255	0.004022140229	2.7×10^{-11}
g_{pq} , X2C	$-4.430966876 \times 10^{-5}$	$-4.430958711 \times 10^{-5}$	8.2×10^{-11}
g_{pq} , 4-component	$-1.946964702 \times 10^{-5}$	$-1.946965256 \times 10^{-5}$	5.5×10^{-12}

(for the four-component run the indices are shifted by the number of negative-energy states, (22, 24) and (23, 25)).

Remark. Occupations must be genuinely fractional on both orbitals of a tested pair, and the pair must have a nonzero gradient: for water the default frontier pair is symmetry-forbidden (both sides vanish), which would make the comparison trivially satisfied.